

1. T 相关

Fermi-Dirac 分布

$$① f_F(E) = \frac{1}{1 + \exp(\frac{E - E_F}{kT})}$$

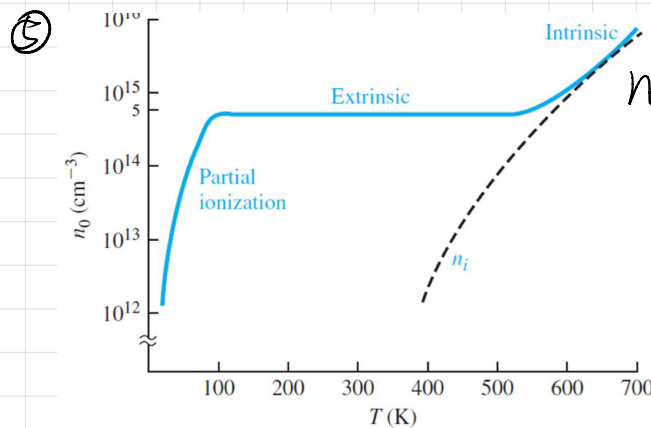
$$|E - E_F| \gg 3kT$$

$$n_0 = N_C \exp \frac{E_F - E_C}{kT}$$

$$② N_C, N_V \propto T^{\frac{3}{2}}$$

$$③ n_i^2 = N_C \cdot N_V \cdot \exp(\frac{-E_g}{kT})$$

$$④ E_F - E_{midgap} = \frac{3}{4} kT \ln \left(\frac{m_p^*}{m_n^*} \right) = \frac{1}{2} kT \ln \frac{N_V}{N_C}$$



$$n = \sqrt{\frac{1}{2} N_C N_D} \cdot \exp \frac{-E_g}{2kT}$$

$$n_d = N_D \cdot \frac{1}{1 + \frac{1}{2} \exp \frac{E_D - E_F}{kT}}$$

$$p_a = N_A \cdot \frac{1}{1 + \frac{1}{2} \exp \frac{E_F - E_A}{kT}}$$

⑥ 迁移率 $\frac{1}{\mu} = \frac{1}{\mu_1} + \frac{1}{\mu_2}$

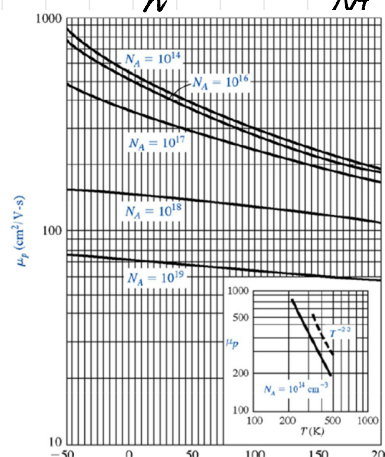
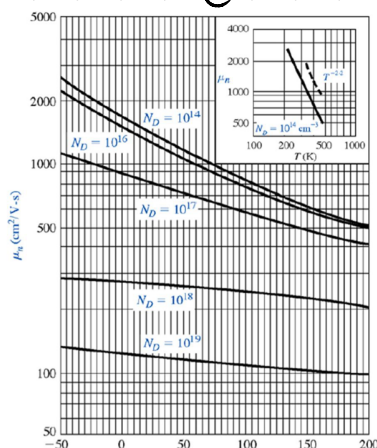
e^-

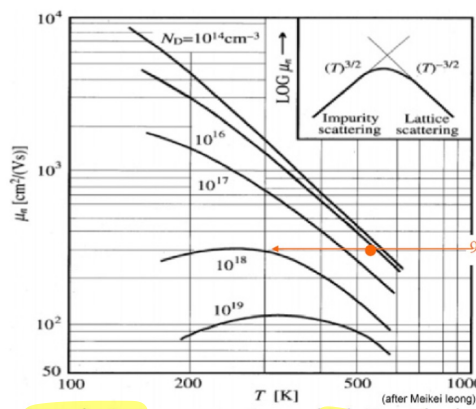
$$\mu_1 \propto T^{-\frac{3}{2}}$$

$$h^+ \mu_2 \propto \frac{T^{\frac{3}{2}}}{N_A}$$

低掺杂 (晶格 散射)

低掺杂 高 conc (电离 杂质)

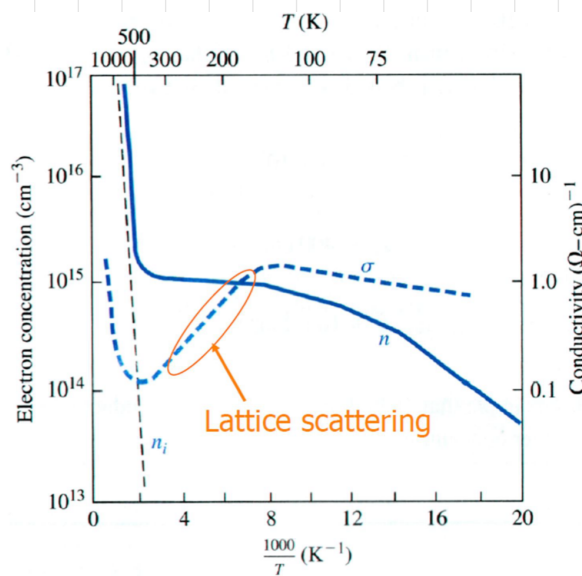




At present doping level, cooling does not improve speed much.

Low doping concentration or high T → The lattice scattering dominates
 High doping concentration or low T → The impurity scattering dominates

⑦



电导率

$$\sigma = q(u_n n + u_p p)$$

2. 浓度相关

① 态密度、浓度

$$n_{(E)} = \int_0^E g_{(E)} \cdot f_{(E)} dE = N_c \cdot \exp \frac{E_f - E_c}{kT} \dots$$

$\int g(E) dE \rightarrow$ 态数目

② 掺杂

$$\begin{array}{lll} T \approx 0K & n_d = N_D & E_f > E_D \\ T > 0K & \text{当 } E_D = E_f & n_d = \frac{2}{3} N_D \end{array}$$

室温

$$n_d \approx 0 \quad E_D > E_f$$

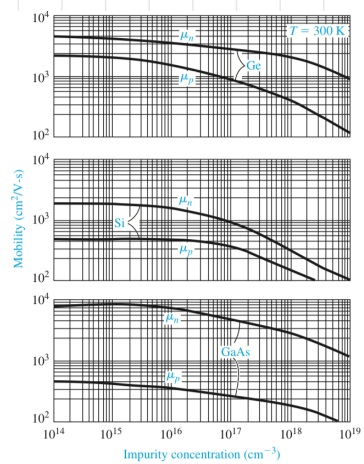
高温

$$E_f = E_{fi} \quad (n = n_i)$$

③ n, p 掺杂

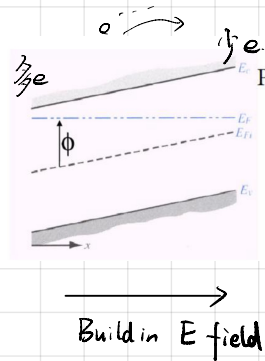
$$n_0 + N_A = p_0 + N_D$$

$$n_0 p_0 = n_i^2 \Rightarrow n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$$



④ $N_d(N_a) \uparrow$, $n_n(n_p) \downarrow$

⑤ 扩散电流



e 能量低,
电势高

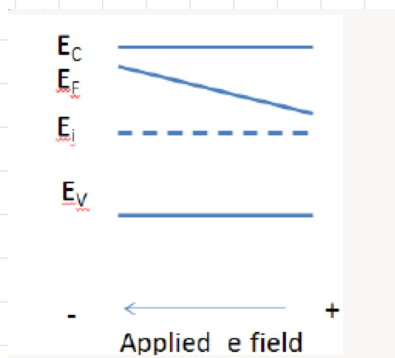
$$eD_n \frac{dn}{dx} - eD_p \frac{dp}{dx}$$

$$E_F \bar{E}_F = kT \ln \frac{N_d N_a}{n_i^2}$$

$$\frac{d\phi}{dx} = - \frac{dE_F}{dx} = \frac{kT}{e} \frac{1}{N_d(x)} \cdot \frac{dN_d}{dx} = \bar{E}_x$$

$$\bar{E}_x = - \frac{kT}{e} \frac{1}{N_d(x)} \frac{dN_d}{dx}$$

外加电场
抵消内建电场



⑥ 平衡

$$n = n_0 + \delta n, p = p_0 + \delta p$$

$$R_0 \propto n_0 \cdot p_0 = \alpha_i \cdot n_i^2 = \frac{n}{\tau_{n0}}$$

$\parallel$
 G_0

⑦ 非平衡半导体

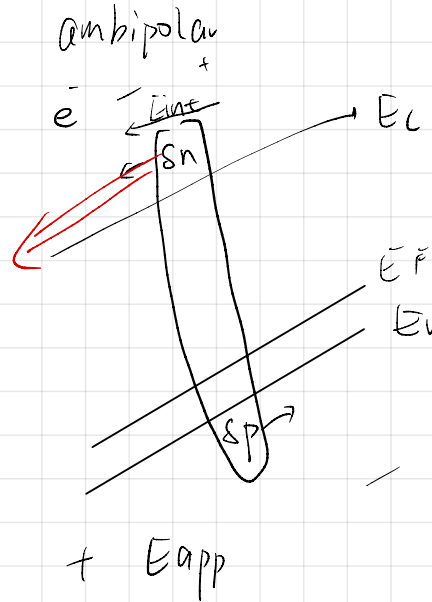
设 p type $p_0 \gg n_0 \Rightarrow \begin{cases} p \approx p_0 \\ n = \delta n \end{cases}$

$$\frac{d(\delta n)}{dt} = \underbrace{\alpha_r \cdot n_i^2}_{产生} - \underbrace{\alpha_r n_0 p_0}_{复合} = -\alpha_r p_0 \cdot \delta n \quad \text{少子决定}$$

$\Rightarrow \delta n(t) = \delta n(0) \cdot e^{-\frac{t}{\tau_{no}}}$
 其中 $\tau_{no} = \frac{1}{\alpha p_0}$, 饱和时 $\delta n = g \tau_{no}$

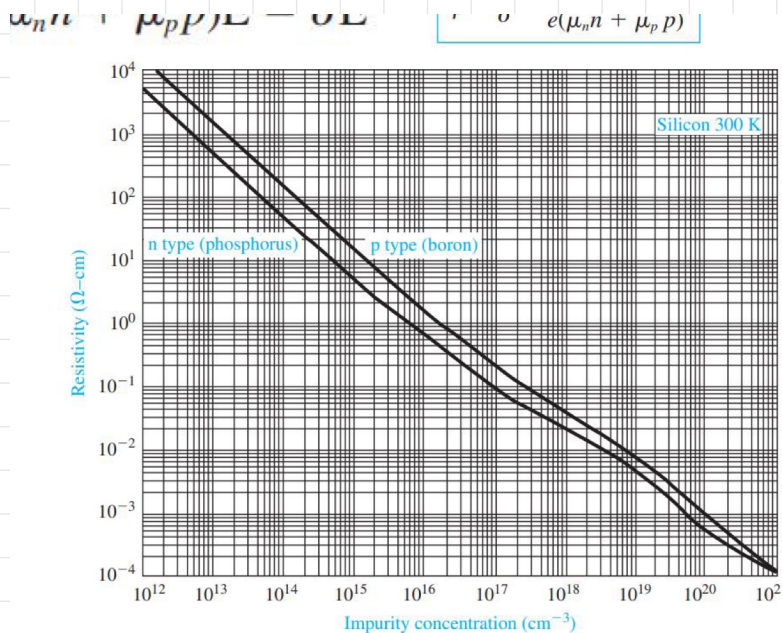
⑧ 双极输运

少数决定载流子



对 hccp $\mu' = \mu_n$
 $D' = D_n = \frac{kT}{e} \mu_n$

⑨



⑩

recombination

generation

diffusion

drift

R_0

G_0

∇n

E

PN结

正偏

反偏

$\sigma(D_n \frac{dn}{dx} - D_p \frac{dp}{dx}) + e(n\mu_n E_x + p\mu_p E_x)$

正偏

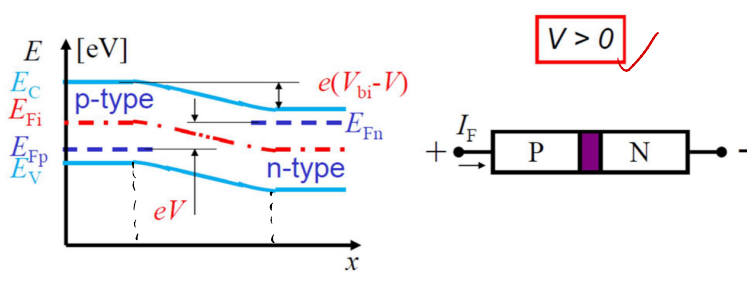
反偏

二.

PN 结

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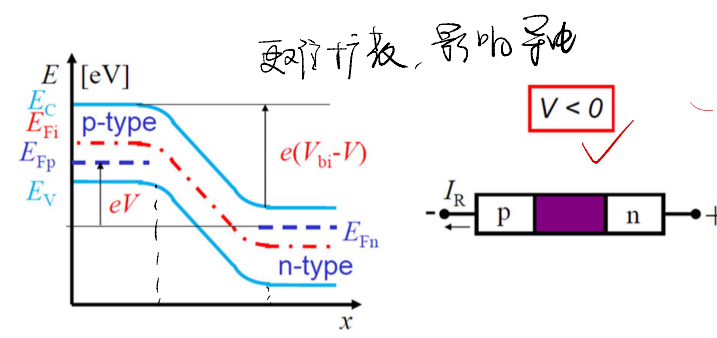
正向 Forward-biased junction



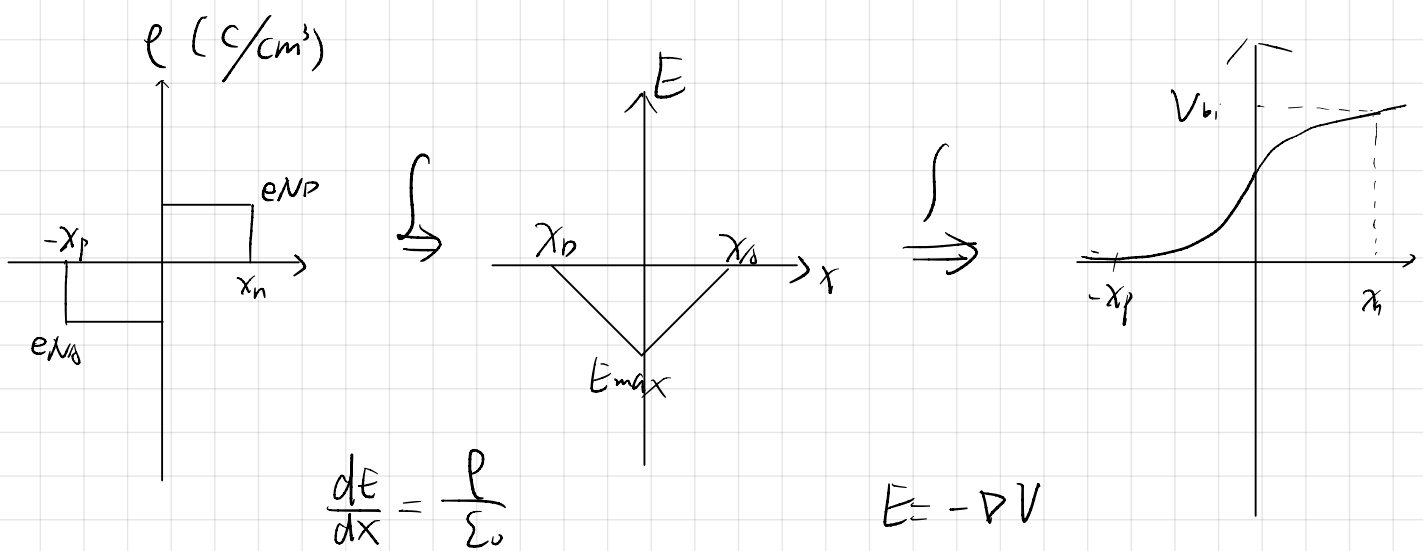
E -field across depletion region decreases.

W decreases, diffusion pushes mobile carriers towards the junction.

反向 Reverse-biased junction



表示出空间电荷区的宽度



Home Work 6-9

对于 PN 结，掺杂浓度为 N_a 和 N_d 。

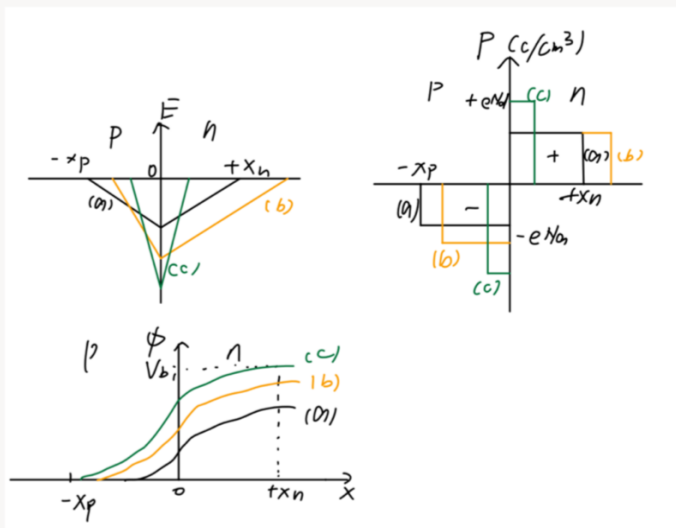
- (a) 画出电荷、电场和电势分布。
- (b) 若 N_a 增加，重复 (a)。
- (c) 若 N_a 和 N_d 都增加，重复 (a)，并比较三者差异。
- (d) 比较这三个结的击穿电压。

- (a)
 - 电荷分布：p区均匀负电荷，n区均匀正电荷，阶跃式。
 - 电场分布：三角形，从 $-x_p$ 增至 $x=0$ 最大，再减至 x_n 为零。
 - 电势分布：抛物线，从 V_{bi} 减至零。
- (b)
 - 电荷分布：p区负电荷密度增加。
 - 电场分布：峰值升高，宽度减小。
 - 电势分布：梯度更陡，总电势差增大。
- (c)
 - 电荷分布：p区和n区电荷密度均增加。
 - 电场分布：峰值更高，宽度更窄。
 - 电势分布：梯度更陡，差异见图。

(d) 比较击穿电压

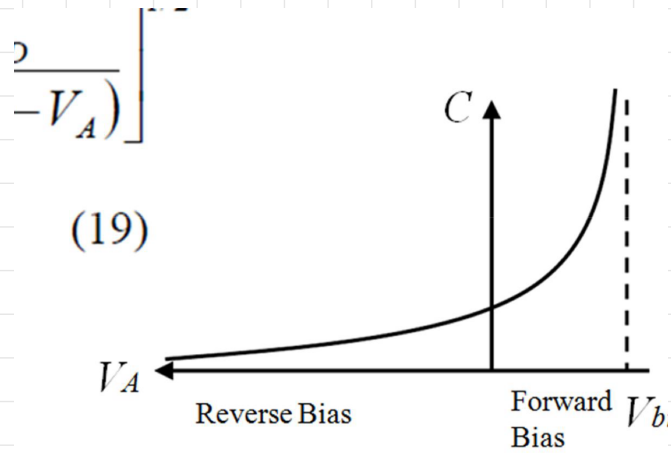
1. 击穿电压分析：P 188 式 7.61

- 击穿电压 $V_{BR} \propto \frac{e|E_{crit}|^2}{qN_a}$ (近似)， $|E_{crit}|$ 为临界电场。
- (a) 初始 N_a 和 N_d ， V_{BR} 较高。
- (b) N_a 增加， V_{BR} 减小，因 N_a 增大，空间电荷区变窄，击穿电场更容易达到。
- (c) N_a 和 N_d 增加， V_{BR} 进一步减小，因 $N_a N_d$ 效应更强。



Depletion capacitance

$$C' = \frac{k \epsilon_0}{W}$$

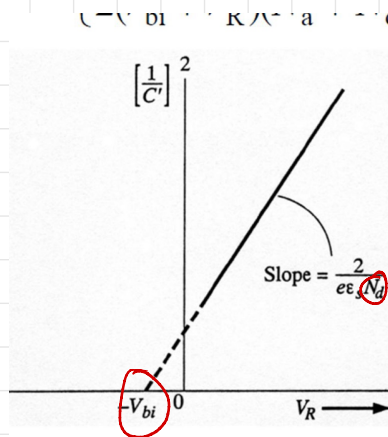


One-side junction

$$(p^+n) \Rightarrow x_p \ll x_n$$

$$W \approx x_n$$

$$C' = \sqrt{\frac{e \epsilon_s N_d}{2(V_{bi} + V_R)}}$$



Breakdown

{

avalanche breakdown

(low doping)

($T \uparrow, V_B \downarrow$) 高电场

Zener (Tunneling)

(high doping)

($T \uparrow, V_B \uparrow$)

低电场

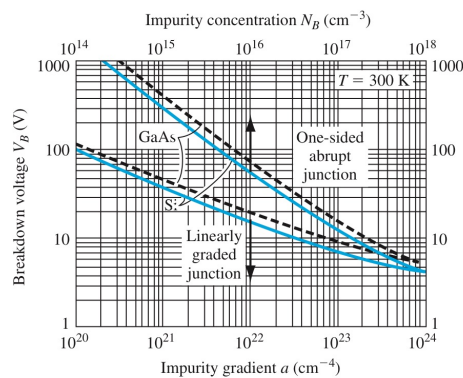
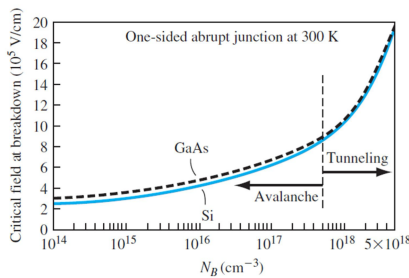
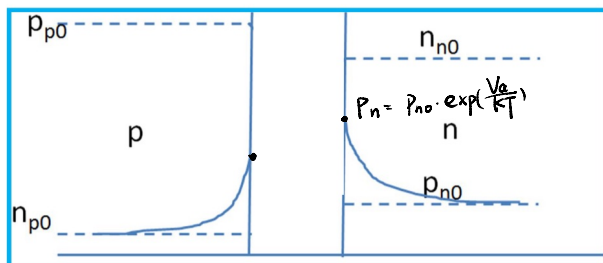


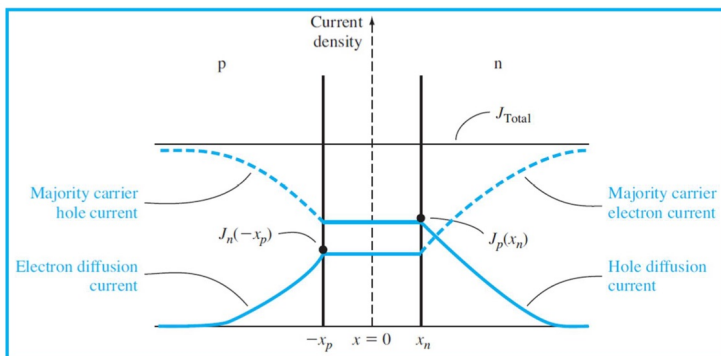
Figure 7.15 | Breakdown voltage versus impurity concentration in uniformly doped and linearly graded junctions. (From Sze [14].)

$$p_n = p_{n0} \exp\left(-\frac{eV_n}{kT}\right) \cdot \exp\left(\frac{eV_a}{kT}\right)$$

Carrier distribution under forward bias



Current distribution under forward bias



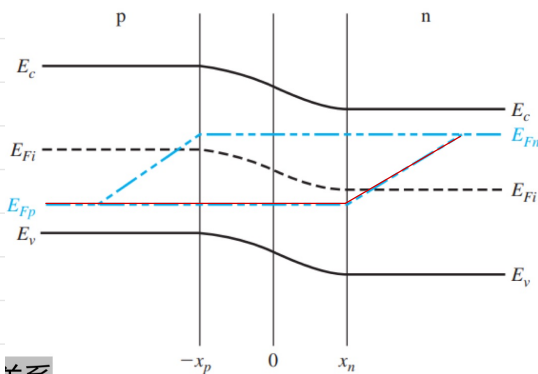
离中心电荷正区

少子的 $J_{diff} \downarrow$, 需要多子弥补

由于多子几乎没有浓度梯度, 由 J_{diff} 驱动

$$J = J_n + J_p = \left(\frac{eD_n \cdot n_{p0}}{L_n} + \frac{eD_p \cdot p_{n0}}{L_p} \right) \left(e^{\frac{eV_a}{kT}} - 1 \right) = J_s \cdot \left(e^{\frac{eV_a}{kT}} - 1 \right)$$

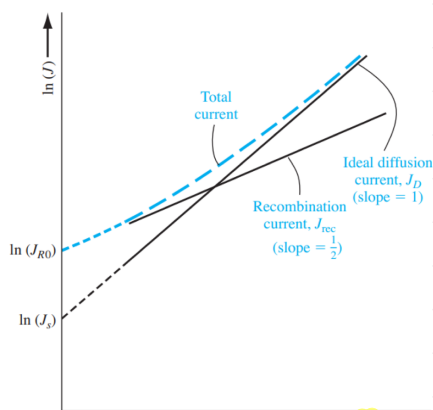
$V_a < 0$ $J \approx -J_s$ 很小(反向)



关系

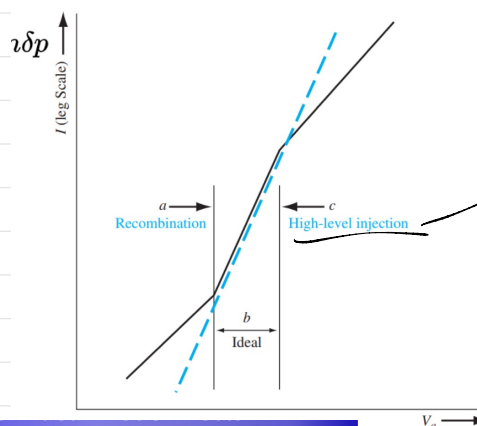
$$J_s = \left[\frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n} \right]$$

$$J = J_s \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$$



$$\frac{eV_a}{kT}$$

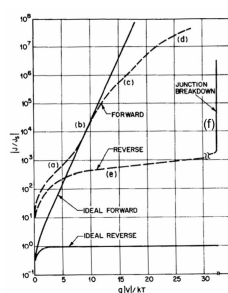
电流密度较低时, 复合电流占主导地位
电流密度较高时, 扩散电流占主导地位



大注入:

正偏电压极高

过剩载流子很多
达到多子量级



- (a) Recombination current
- (b) Diffusion current
- (c) High field injection
- (d) Series resistance effect
- (e) Generation current
- (f) Breakdown current

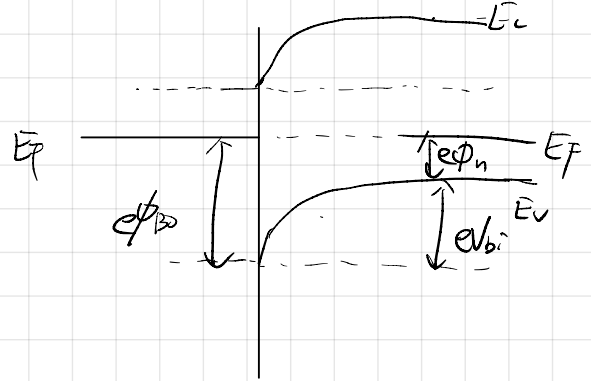
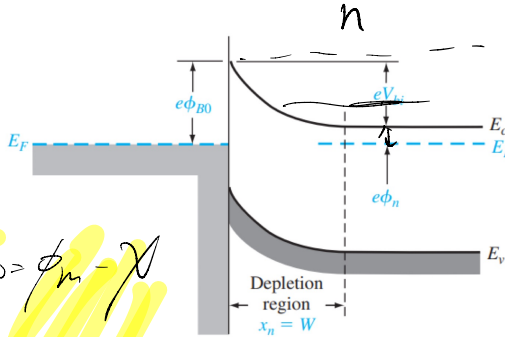
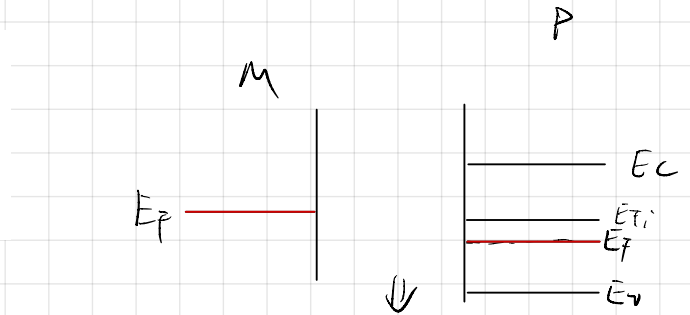
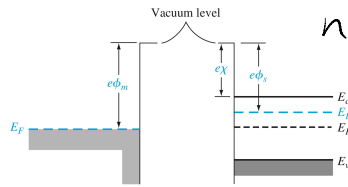
$$n \cdot p = n_i^2 \exp \frac{V_a}{V_t} \approx n_n \cdot p_p$$

$$n_n \approx p_p = n_i \exp \frac{V_a}{2V_t}$$

$$I_{oc} \exp \frac{V_a}{2V_t}$$

$M-S = \text{极结}$

$\phi_n = \phi_s - \chi$



$\phi_{B0} = \phi_m - \chi$

$V_{bi} = \phi_{B0} - \phi_n$

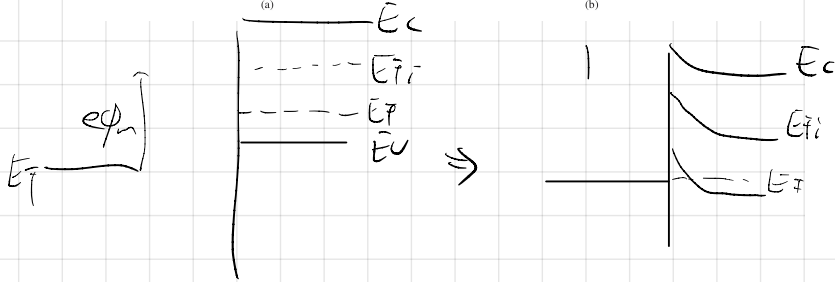
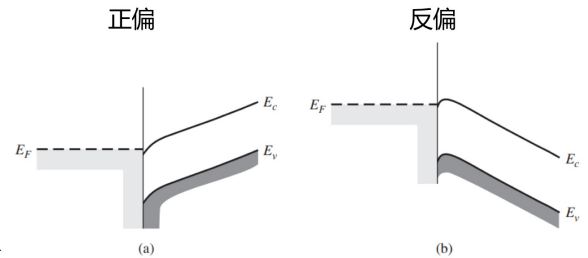
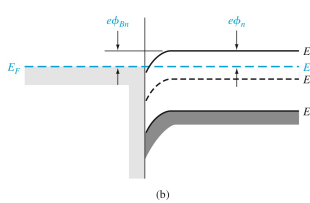
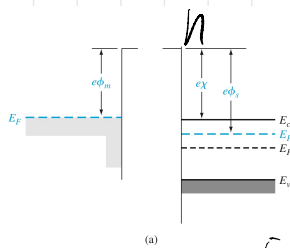
$= \phi_m - \phi_s$

$J_{sT} = A^* T^2 \exp\left(\frac{-e\phi_{Bn}}{kT}\right)$

$J = J_{sT} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$

欧姆接触

- A) Dope the semiconductor heavily enough that tunneling is possible.
- B) Choose a system where ϕ_M and ϕ_S are such that there is no barrier to current flow



欧姆接触:

n型半导体: 金属功函数小于半导体功函数

p型半导体: 金属功函数大于半导体功函数

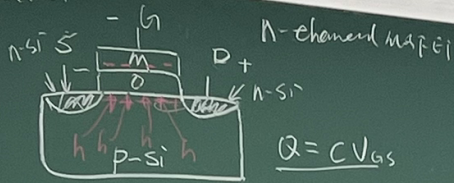
反的

表面态

肖特基效应

n:

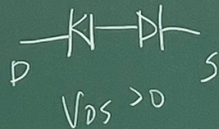
accumulation



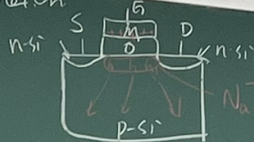
$$V_G = 0, V_{DS} = 0, I_{DS} = 0$$

$$V_{GS} = 0, V_{DS} > 0, \bar{n} \rightarrow p \rightarrow \bar{n}, I_{DS} = 0$$

$$V_{GS} < 0, V_{DS} > 0, \bar{n} \rightarrow p \rightarrow \bar{n}, I_{DS} = 0$$



depletion



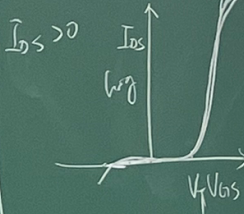
$$0 < V_{GS} < V_T, \bar{n} \rightarrow \bar{n}, I_{DS} = 0$$

$$V_{DS} > 0,$$

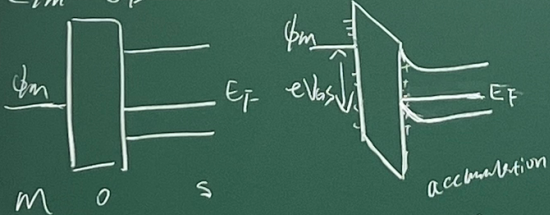
inversion



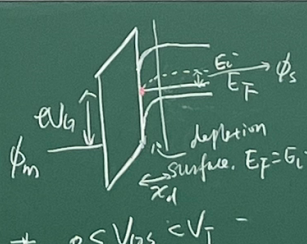
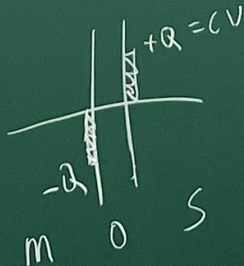
$$0 < V_{GS} = V_T, V_{DS} > 0, \bar{n} \rightarrow \bar{n}$$



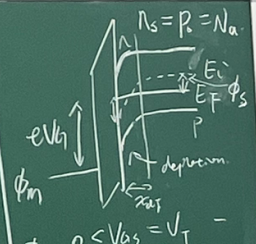
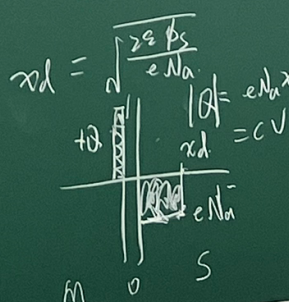
$$e\phi_m = E_F$$



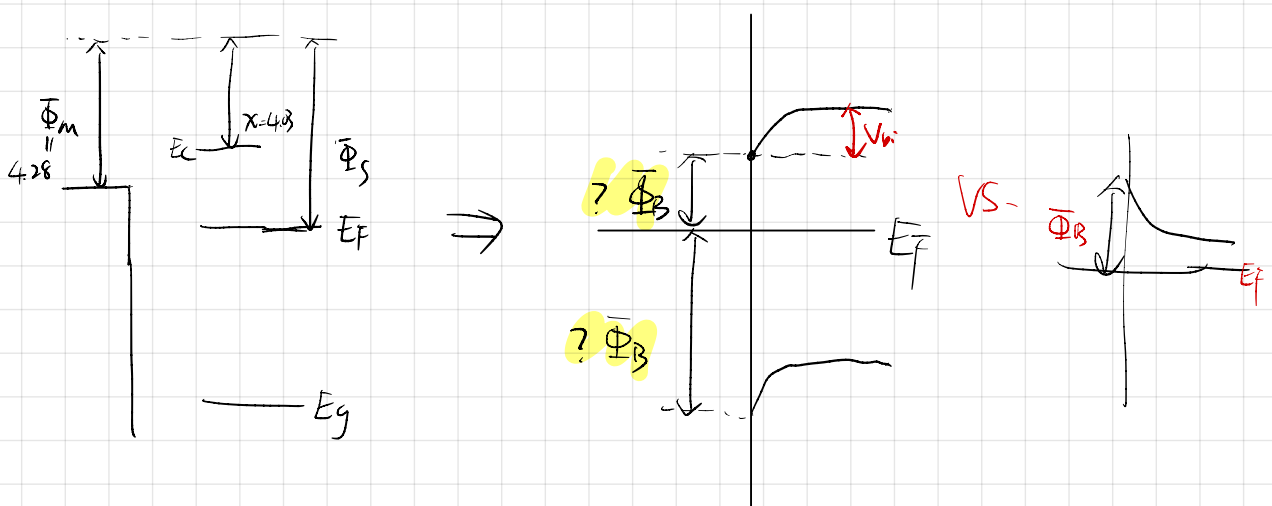
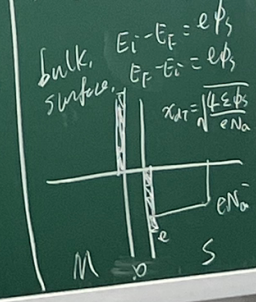
$$V_{GS} = 0, V_{GS} < 0 +$$



$$0 < V_{GS} < V_T -$$



$$0 < V_{GS} = V_T -$$



$$E_C - E_F = kT \ln \frac{N_A}{n_i} = 0.41 \text{ eV}$$

$$\Phi_S = 4.44 \text{ eV} > \Phi_m$$